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Application of the Deep Learning Integrated Framework CEEMDAN-GRU-Informer in Financial Time Series Prediction

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challenges associated with financial time series prediction by proposing the CEEMDAN-GRU-Informer model within a deep learning framework. By incorporating the latest Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) technology, the original financial time series is decomposed to extract price variation information across different time scales. The complexity of each decomposed sequence is assessed using the Sample Entropy (SE) method, facilitating the selection of an appropriate predictive model. In the model prediction phase, a hybrid approach combining Gated Recurrent Unit (GRU) and Informer architectures is employed to enhance forecasting accuracy, leveraging various data-driven techniques to adapt to the complexity of the sequences. This research aims to significantly improve forecasting capabilities, offering stakeholders valuable insights for decision-making in volatile and dynamic financial environments. Empirical studies demonstrate that, within the SZ1 and SZ6 datasets, the root mean square error (RMSE) of the proposed model is 0.0007 and 0.0009, respectively, indicating substantial improvements over other hybrid models and highlighting its

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References

Bustos, O., & Pomares, A. (2020). Stock market movement forecast: A Systematic review. *Expert Systems with Applications*, 156, 113464. <https://doi.org/10.1016/j.eswa.2020.113464>

[Article](#) [Google Scholar](#)

Caraiani, P. (2020). Forecasting financial networks. *Computational Economics*, 55, 983–997. <https://doi.org/10.1007/s10614-019-09925-8>

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Ding, G., & Qin, L. (2020). Study on the prediction of stock price based on the associated network model of LSTM. *International Journal of Machine Learning and Cybernetics*, 11, 1307–1317. <https://doi.org/10.1007/s13042-019-01041-1>

Haan, L., Mercadier, C., & Zhou, C. (2016). Adapting extreme value statistics to financial time series: Dealing with bias and serial dependence. *Finance and Stochastics*, 20, 321–354. <https://doi.org/10.1007/s00780-015-0287-6>

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Lin, Z., & Ha-Kyun, K. (2020). The influence of financial service characteristics on use intention through customer satisfaction with mobile fintech. *Journal of System and Management Sciences*, 10, 82–94. <https://doi.org/10.33168/JSMS.2020.0206>

[Article](#) [Google Scholar](#)

Lu, W., Li, J., Li, Y., Sun, A., & Wang, J. (2020). A CNN-LSTM-based model to forecast stock prices. *Complexity*, 6, 1–10. <https://doi.org/10.1155/2020/6622927>

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Niu, H., Xu, K., & Wang, W. (2020). A hybrid stock price index forecasting model based on variational mode decomposition and LSTM network. *Applied Intelligence*, 50, 4296–4309. <https://doi.org/10.1007/s10489-020-01814-0>

[Article](#) [Google Scholar](#)

Nti, K., Adekoya, A., & Weyori, B. (2020). A systematic review of fundamental and technical analysis of stock market predictions. *Artificial Intelligence Review*, 53, 3007–3057. <https://doi.org/10.1007/s10462-019-09754-z>

[Article](#) [Google Scholar](#)

Staffini, A. (2022). Stock price forecasting by a deep convolutional generative adversarial network. *Frontiers in Artificial Intelligence*, 5, 837596.

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Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A., et al. (2017). Attention is all you need. *Proceedings of the 31st International Conference on Neural Information Processing Systems*, 17, 6000–6010. <https://doi.org/10.48550/arXiv.1706.03762>

[Article](#) [Google Scholar](#)

Wang, J., & Wang, J. (2017). Forecasting stochastic neural network based on financial empirical mode decomposition. *Neural Networks*, 90, 8–20. <https://doi.org/10.1016/j.neunet.2017.03.004>

[Article](#) [Google Scholar](#)

Wang, J., Feng, L., Li, Y., He, J., & Feng, C. (2021). Deep nonlinear ensemble framework for stock index forecasting and uncertainty analysis. *Cognitive Computation*, 13, 1574–1592.

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Yilmaz, F., & Yildiztepe, E. (2024). Statistical evaluation of deep learning models for stock return forecasting. *Computational Economics*, 63, 221–244. <https://doi.org/10.1007/s10614-022-10338-3>

[Article](#) [Google Scholar](#)

Yu, X., & Huang, Y. (2021). The impact of economic policy uncertainty on stock volatility: Evidence from GARCH–MIDAS approach. *Physica A: Statistical Mechanics and Its Applications*, 570, 125794. <https://doi.org/10.1016/j.physa.2021.125794>

[Article](#) [Google Scholar](#)

Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., et al. (2021). Informer: Beyond efficient transformer for long sequence time-series forecasting. *Proceedings of the AAAI*

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